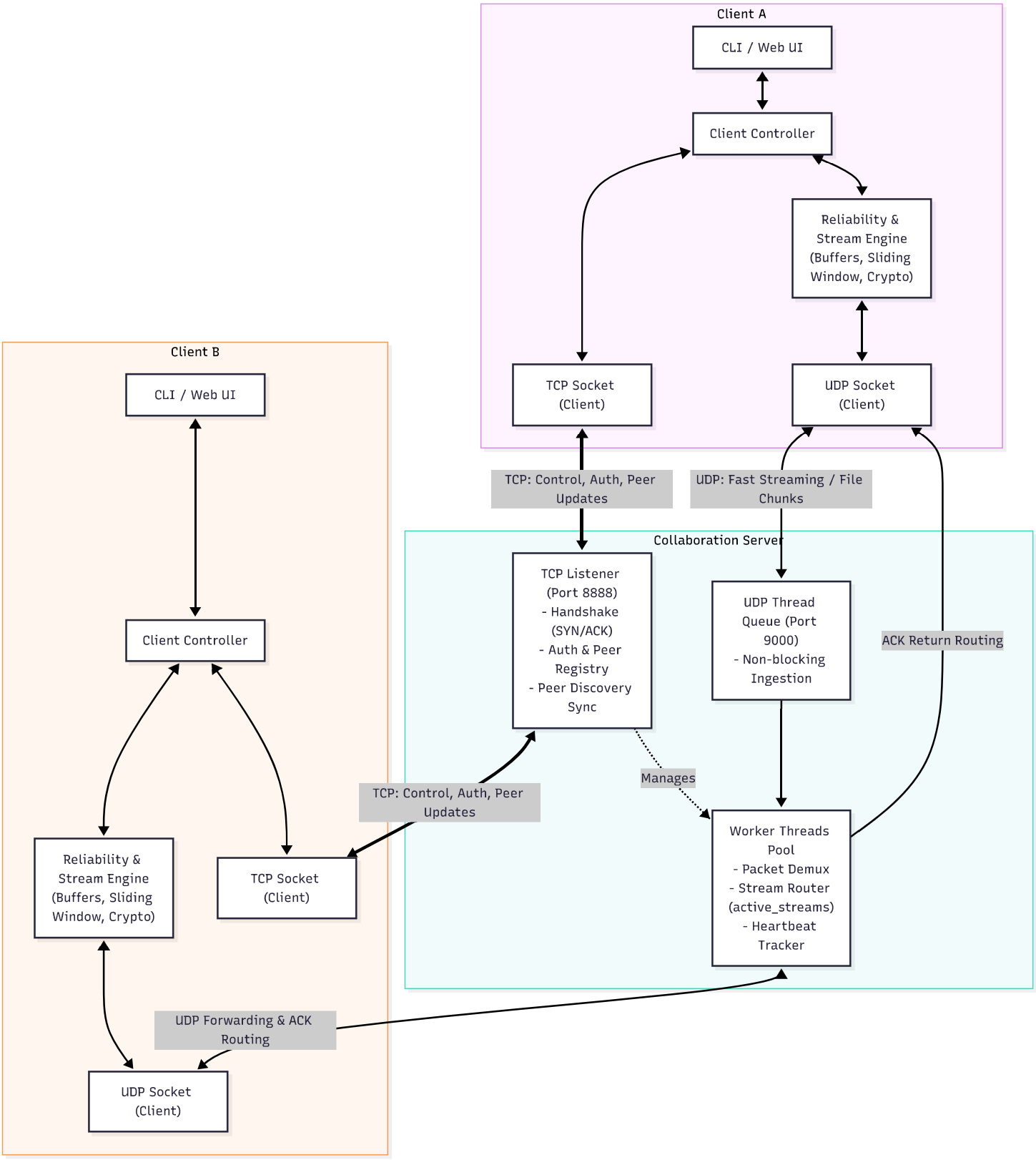

Client A
CLI / Web UI
Client Controller
Reliability & Stream Engine (Buffers, Sliding Window, Crypto)
TCP Socket (Client)
UDP Socket (Client)
Client B
CLI / Web UI
Client Controller
Reliability & Stream Engine (Buffers, Sliding Window, Crypto)
TCP Socket (Client)
UDP Socket (Client)
TCP: Control, Auth, Peer Updates
UDP: Fast Streaming / File Chunks
Collaboration Server
TCP Listener (Port 8888) - Handshake (SYN/ACK) - Auth & Peer Registry - Peer Discovery Sync
UDP Thread Queue (Port 9000) - Non-blocking Ingestion
Manages
Worker Threads Pool - Packet Demux - Stream Router (active_streams) - Heartbeat Tracker
ACK Return Routing
TCP: Control, Auth, Peer Updates
UDP Forwarding & ACK Routing